EE5334 Assignment 1

Objective 2: Cepstral Analysis of Telugu Vowels

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Abstract

Cepstral analysis was performed on three recorded Telugu vowels: Amma (/a/), Illu (/i/), and Uduta (/u/). For each vowel, eight consecutive 25 ms frames with a 5 ms frame shift were selected from the highest-energy block within the segmented vowel. The real cepstrum was used to estimate pitch, while low-quefrency liftering with $K = 30$ was used to obtain a smoothed spectral envelope for F1, F2, and F3 estimation.

The average pitch estimates were 159.99 Hz for Amma, 149.90 Hz for Illu, and 151.55 Hz for Uduta. Framewise results show stable pitch across the selected frames, while some formant estimates vary more strongly, particularly the F2 estimate in the first Amma and Uduta frames.

1 Objective

The second objective requires estimation of average pitch and the first three formant frequencies of a recorded vowel using at least six consecutive frames. It also requires framewise plots of the cepstrally smoothed spectra and cepstral sequences. The implementation analysed eight consecutive frames for each of the three selected Telugu vowels, thereby exceeding the minimum frame-count requirement.

2 Speech Material and Frame Selection

The vowel recording `Telugu_Vowels.wav` was read in MATLAB and resampled to 10 kHz. Pre-emphasis with coefficient 0.8 was applied before cepstral processing. Three vowel tokens were analysed: Amma (/a/), Illu (/i/), and Uduta (/u/). The approximate vowel boundaries and the selected frame-centre times are listed in Table 1.

Table 1: Vowel boundaries and centre times of the eight selected frames.

Word	Vowel interval (s)	First to last frame centre (s)	Frame-centre times (s)
Amma (/a/)	1.77–1.94	1.812–1.847	1.812, 1.817, 1.822, 1.827, 1.832, 1.837, 1.842 1.847
Illu (/i/)	4.83–5.00	4.842–4.877	4.842, 4.847, 4.852, 4.857, 4.862, 4.867, 4.872 4.877
Uduta (/u/)	8.22–8.39	8.297–8.332	8.297, 8.302, 8.307, 8.312, 8.317, 8.322, 8.327 8.332

For each vowel, the mean-square energy was calculated for every possible frame. The average energy of each possible block of eight consecutive frames was then compared, and the block with the highest average energy was selected. The frame-centre times in Table 1 specify the locations of the eight analysis frames.

Since the frame duration is 25 ms and the shift is 5 ms, the total interval from the beginning of the first frame to the end of the eighth frame is

$$T_{\text{block}} = 25 + 7(5) = 60 \text{ ms}. \tag{1}$$

This energy-based selection places the analysis block over the stronger part of the vowel rather than using an arbitrary fixed position in the segment.

3 Analysis Parameters

The cepstral-analysis parameters are summarised in Table 2.

Table 2: Task 2 cepstral-analysis parameters.

Parameter	Setting
Analysis sampling rate	10 kHz
Pre-emphasis coefficient	0.8
Frame duration	25 ms
Frame shift	5 ms
Number of consecutive frames	8
Window	Hamming
FFT length	8192
Cepstral lifter parameter	$K = 30$
Pitch search range	60–350 Hz
Minimum F1–F2 separation	300 Hz
Minimum F2–F3 separation	450 Hz

At a sampling frequency of 10 kHz, one cepstral sample corresponds to 0.1 ms. Retaining coefficients from quefrency index 0 through $K = 30$ therefore retains the low-quefrency region up to approximately

$$q_{\max} = \frac{K}{F_s} = \frac{30}{10000} = 0.003 \text{ s} = 3 \text{ ms}. \tag{2}$$

The matching coefficients at the end of the circular cepstral sequence are also retained to preserve the corresponding negative-quefrency portion during reconstruction.

4 Cepstral Analysis Method

4.1 Real Cepstrum

For a windowed speech frame $x_w[n]$, the FFT is calculated and the logarithm of its magnitude is taken. The real cepstrum is then

$$c[n] = \mathcal{F}^{-1} \left\{ \ln \left(|X_w[k]| + \epsilon \right) \right\}, \tag{3}$$

where ϵ prevents the logarithm of zero. The cepstral index is interpreted as quefrency. The pitch period appears as a peak in the real cepstrum within the expected quefrency interval.

4.2 Pitch Estimation

The cepstral peak is searched over periods corresponding to the selected pitch limits of 60–350 Hz. If the peak occurs at quefrency τ_0, the fundamental frequency is

$$f_0 = \frac{1}{\tau_0}. \tag{4}$$

Parabolic interpolation using the peak and its two neighbouring cepstral samples refines the quefrency estimate. The average pitch for each vowel is calculated from its eight framewise estimates.

4.3 Cepstral Liftering

The low-quefrency cepstral coefficients describe the slowly varying spectral envelope, while higher-quefrency coefficients contain much of the fine harmonic structure. A low-time lifter retains

coefficients from index 0 through $K = 30$ and the matching coefficients at the end of the circular cepstral sequence. The remaining coefficients are set to zero.

The smoothed spectrum is reconstructed as

$$\widehat{S}[k] = \exp\left(\mathcal{F}\{c_K[n]\}\right), \tag{5}$$

where $c_K[n]$ is the liftered cepstrum. The reconstructed envelope reduces fine harmonic ripple and makes the broader spectral maxima easier to inspect.

4.4 Formant Selection and Minimum Separation

The strongest local envelope maximum is selected within a broad vowel-dependent range for F1. The lower search boundary for F2 is then raised to at least $F_1 + 300$ Hz. Similarly, the lower F3 boundary is raised to at least $F_2 + 450$ Hz:

$$F_2 - F_1 \geq 300\,\text{Hz}, \tag{6}$$
$$F_3 - F_2 \geq 450\,\text{Hz}. \tag{7}$$

These separations reduce the likelihood that two nearby local maxima will be assigned as consecutive formants. They are peak-selection constraints and should not be interpreted as universal physiological formant-spacing rules. If the true resonance is weak or broad, the constrained peak search can select another local maximum; consequently, the smoothed-envelope plots remain important for interpreting the numerical estimates.

5 Average Results

The average pitch and formant estimates over the eight selected frames are summarised in Table 3.

Table 3: Average cepstral estimates for the three Telugu vowels.

Word / vowel	Average pitch (Hz)	Average F_1 (Hz)	Average F_2 (Hz)	Average F_3 (Hz)
Amma (/a/)	159.99	932.31	2050.17	3573.30
Illu (/i/)	149.90	370.03	2962.19	3592.99
Uduta (/u/)	151.55	481.26	1024.32	2032.47

6 Framewise Results

6.1 Amma (/a/)

Table 4: Framewise cepstral estimates for Amma (/a/).

Frame	Centre time (s)	F_0 (Hz)	F_1 (Hz)	F_2 (Hz)	F_3 (Hz)
1	1.812	160.36	944.8	2751.5	3591.3
2	1.817	159.98	959.5	1992.2	3606.0
3	1.822	159.22	953.4	1805.4	3606.0
4	1.827	159.00	955.8	2003.2	3581.5
5	1.832	159.55	952.1	1984.9	3586.4
6	1.837	159.18	922.9	1959.2	3520.5
7	1.842	160.70	896.0	1991.0	3522.9
8	1.847	161.93	874.0	1914.1	3571.8

The Amma pitch remains close to 160 Hz throughout the selected frames. F1 decreases gradually from 944.8 Hz in the first frame to 874.0 Hz in the eighth frame. F3 remains relatively stable, between 3520.5 and 3606.0 Hz.

F2 has one pronounced high value of 2751.5 Hz in the first frame, whereas frames 2–8 lie between 1805.4 and 2003.2 Hz. This first-frame value increases the arithmetic mean F2 to 2050.17 Hz. The framewise spectrum should therefore be inspected when interpreting the average F2; the mean alone does not show the isolated first-frame deviation.

6.2 Illu (/i/)

Table 5: Framewise cepstral estimates for Illu (/i/).

Frame	Centre time (s)	F_0 (Hz)	F_1 (Hz)	F_2 (Hz)	F_3 (Hz)
1	4.842	149.88	367.4	2945.6	3568.1
2	4.847	149.54	341.8	2976.1	3581.5
3	4.852	149.32	350.3	2973.6	3593.8
4	4.857	149.56	378.4	2974.9	3586.4
5	4.862	149.76	385.7	2982.2	3598.6
6	4.867	150.07	379.6	2971.2	3621.8
7	4.872	150.29	371.1	2960.2	3593.8
8	4.877	150.77	385.7	2913.8	3599.9

For Illu, pitch is particularly stable, varying from 149.32 to 150.77 Hz. F1 remains between 341.8 and 385.7 Hz. F2 is also stable, between 2913.8 and 2982.2 Hz, and F3 remains between 3568.1 and 3621.8 Hz.

Of the three vowels, Illu shows the most consistent framewise formant estimates in this selected block. Its low F1 and high F2 are consistent with the expected acoustic pattern of a high front vowel.

6.3 Uduta (/u/)

Table 6: Framewise cepstral estimates for Uduta (/u/).

Frame	Centre time (s)	F_0 (Hz)	F_1 (Hz)	F_2 (Hz)	F_3 (Hz)
1	8.297	149.97	544.4	1667.5	2292.5
2	8.302	150.93	577.4	877.7	2232.7
3	8.307	151.53	515.1	815.4	2154.5
4	8.312	152.21	456.5	883.8	2143.6
5	8.317	153.12	446.8	994.9	1969.0
6	8.322	153.02	434.6	968.0	1894.5
7	8.327	151.86	423.6	974.1	1812.7
8	8.332	149.77	451.7	1013.2	1760.3

Uduta pitch remains within 149.77–153.12 Hz. Its formant estimates vary more than those of Illu. The first-frame F2 value is 1667.5 Hz, while frames 2–8 are between 815.4 and 1013.2 Hz. F3 also decreases across the selected sequence, from 2292.5 Hz in frame 1 to 1760.3 Hz in frame 8.

F1 decreases from 577.4 Hz in frame 2 toward values around 423–452 Hz in the later frames. These changes show that the spectral envelope is not equally stationary across the selected block. They may reflect a changing vocal-tract configuration within the vowel segment, peak-selection sensitivity, or both; the smoothed spectra are needed to distinguish these possibilities.

7 Comparison of Pitch Estimates Across the Two Objectives

The Task 1 harmonic-spacing and AMDF estimates can be compared with the Task 2 average cepstral pitch. Table 7 shows that all three methods give similar values for the corresponding vowel tokens.

Table 7: Pitch estimates obtained using the three methods.

Vowel	Task 1 harmonic spacing (Hz)	Task 1 AMDF (Hz)	Task 2 cepstral average (Hz)
Amma (/a/)	160.00	158.89	159.99
Illu (/i/)	150.75	152.36	149.90
Uduta (/u/)	148.75	150.49	151.55

The close agreement across the independent pitch methods is an important consistency check. The largest difference between the Task 2 cepstral average and the Task 1 harmonic-spacing estimate is 2.80 Hz for Uduta. The pitch estimates are therefore mutually consistent at the level of the present short-time analysis.

8 Observations and Limitations

The following observations were made from the numerical results and the framewise behaviour:

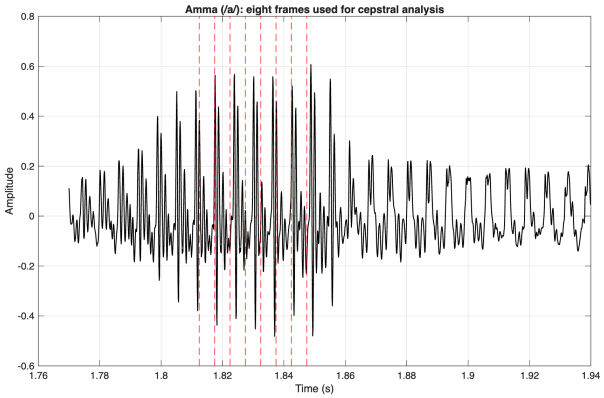
1. **Pitch is stable across the selected blocks.** All three vowels have average pitch between approximately 150 and 160 Hz. The framewise values vary only slightly within each eight-frame sequence.
2. **Illu gives the most stable formant sequence.** Its F1, F2, and F3 estimates change relatively little from frame to frame, and its high F2 distinguishes it from the other two vowels.

3. **Amma has an F2 outlier in the first frame.** The first-frame estimate of 2751.5 Hz is well above the remaining seven values. This illustrates why framewise plots are required in addition to average values.
4. **Uduta has noticeable formant movement.** F2 and F3 decrease over the selected frames, while F1 also changes. The average values summarise the block but do not represent every frame equally well.
5. **The minimum-separation rule avoids adjacent peak assignments.** The 300 Hz F1–F2 and 450 Hz F2–F3 gaps reduce the selection of neighbouring local maxima as separate formants. These limits improve peak separation but do not guarantee that every selected maximum is the physical formant centre.
6. **Cepstral smoothing depends on the lifter.** With $K = 30$, the retained low-quefrency region extends to approximately 3 ms at 10 kHz. A different K changes the envelope smoothness and may change the positions of its local maxima.

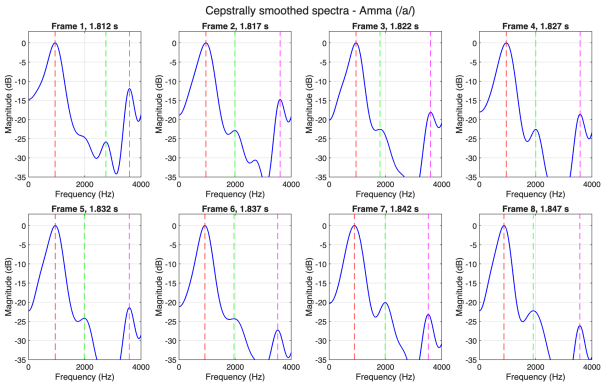
9 Framewise Plots

For each vowel, the figures show the selected waveform and frame centres, the eight cepstrally smoothed spectra, the eight real cepstral sequences, and the average pitch and formant estimates.

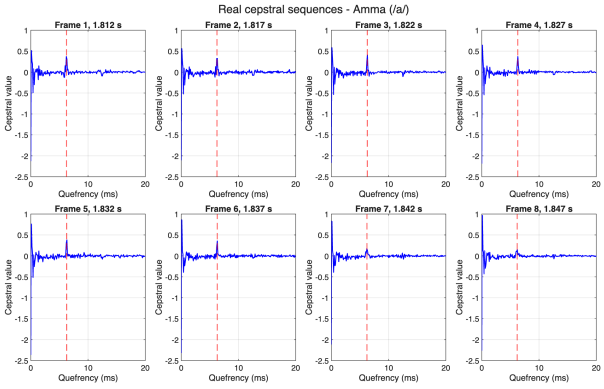
9.1 Amma (/a/)



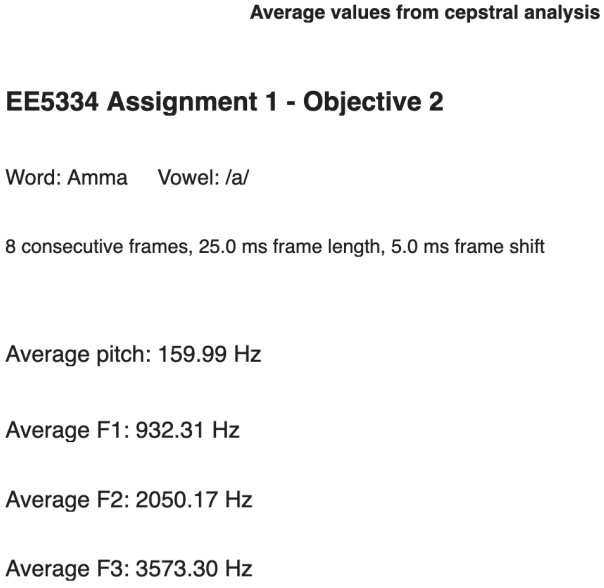
(a) Selected waveform and frame centres.



(b) Cepstrally smoothed spectra.



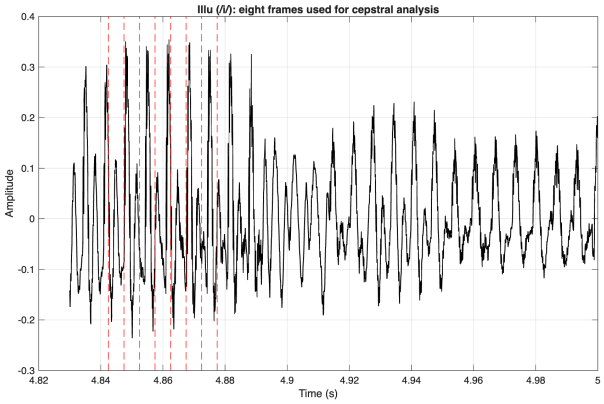
(c) Real cepstral sequences.



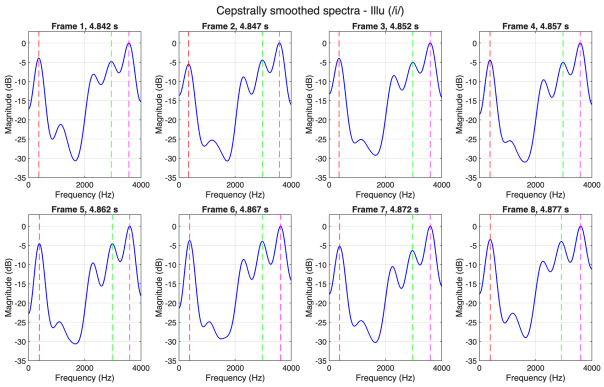
(d) Average pitch and formants.

Figure 1: Task 2 plots for Amma (/a/).

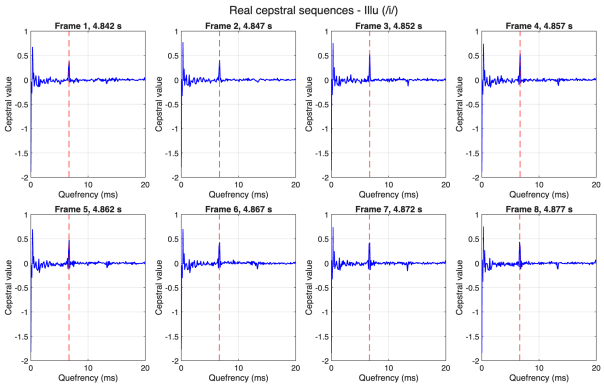
9.2 Illu (/i/)



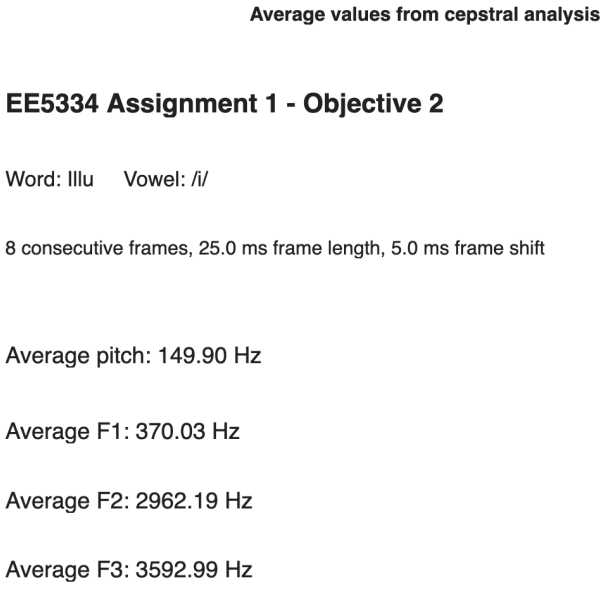
(a) Selected waveform and frame centres.



(b) Cepstrally smoothed spectra.



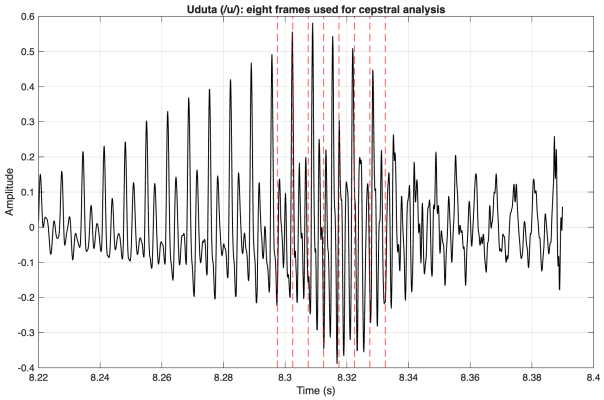
(c) Real cepstral sequences.



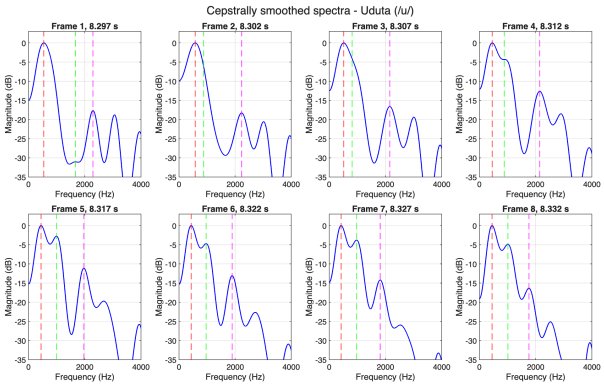
(d) Average pitch and formants.

Figure 2: Task 2 plots for Illu (/i/).

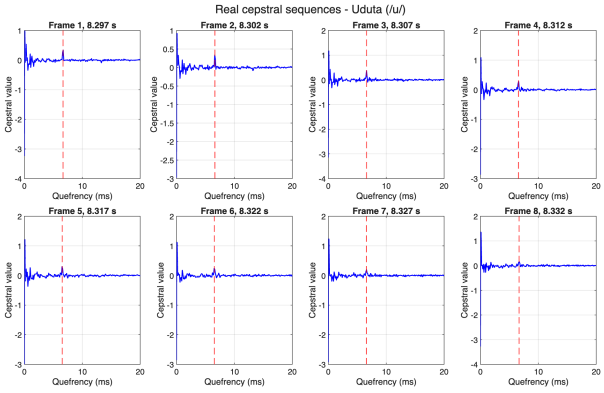
9.3 Uduta (/u/)



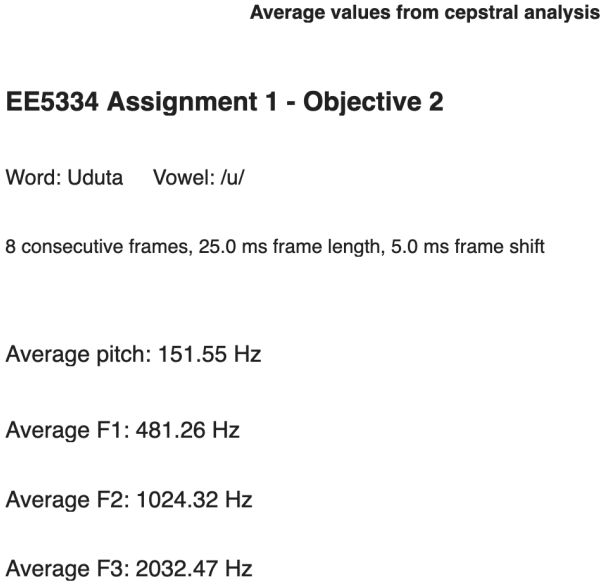
(a) Selected waveform and frame centres.



(b) Cepstrally smoothed spectra.



(c) Real cepstral sequences.



(d) Average pitch and formants.

Figure 3: Task 2 plots for Uduta (/u/).

10 Conclusion

Cepstral analysis was applied to Amma (/a/), Illu (/i/), and Uduta (/u/) using eight consecutive 25 ms frames with a 5 ms shift. The low-quefrency lifter used $K = 30$, corresponding to approximately 3 ms at the 10 kHz analysis rate. Average pitch values were 159.99 Hz, 149.90 Hz, and 151.55 Hz, respectively. These values agree closely with the Task 1 harmonic-spacing and AMDF estimates.

Illu showed the most stable formant estimates across its eight frames. Amma exhibited a high first-frame F2 estimate, and Uduta showed more substantial framewise movement in F2 and F3. The results demonstrate both the usefulness of cepstral smoothing for estimating spectral envelopes and the importance of inspecting framewise plots before interpreting averages as stationary formant values.

References

[1] EE5334, *Assignment 1*, Objective 2: Cepstral analysis of a recorded vowel, course handout, 2026.